Discrete Mathematical Structures
Module 3- Relations & Functions

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S3 DMS MAT 203 (CS, IT)

MODULE-3 RELATIONS & FUNCTIONS

Syllabus.

* Cartesian Product, Binary Relation, Function, Domain, Range, One to One Function Image - Restrictions.
* Properties, Reachability Relations, Reflexive, symmetric transitive relations, Antisymmetric Relations.
* Partial order Relation
* Equivalence Relation, Irreflexive Relations
* Partially ordered set (POSET), Hasse Diagram.
* Maximal- Minimal Element, Least upper bound (LUB), Greatest lower bound (GLB)
* Equivalence Relation & partitions, equivalence class.
* Lattice - Dual lattice, Sublattice, properties of glb & lub.
* Properties of lattice, special lattice, Complete lattice, Bounded lattice, Complimented lattice, Distributive lattice.

I

Cartesian Product (Cross Product)

For sets A, B the cartesian product or cross product of A & B is denoted by $A \times B$ and

$A \times B = \{(a,b) \mid a \in A,\ b \in B\}$

* If A and B are finite $\boxed{|A \times B| = |A| \times |B|}$
* We say that the elements of $A \times B$ are ordered pairs.

Extension of Cartesian product to more than 2 sets.

For sets $A_1, A_2, A_3 \cdots A_n$ the cartesian product

$A_1 \times A_2 \times A_3 \times \cdots \times A_n = \{(a_1, a_2, a_3 \cdots a_n) \mid a_i \in A_i,\ \forall i\}$

* $\boxed{|A_1 \times A_2 \times A_3 \times \cdots \times A_n| = |A_1| \times |A_2| \times |A_3| \times \cdots \times |A_n|}$
* We say that the elements of $A_1 \times A_2 \times \cdots \times A_n$ are called ordered n-tuples

A 3-tuple is also known as a triple.

1) Let $A = \{1,2,3\}$ $B = \{x, y\}$ find $A \times B$, $B \times A$, B^2, B^3

$A \times B = \{(1,x), (1,y), (2,x), (2,y), (3,x)(3,y)\}$

$B \times A = \{(x,1), (x,2), (x,3), (y,1)(y,2), (y,3)\}$

$B^2 = B \times B = \{(x,x), (x,y), (y,x), (y,y)\}$

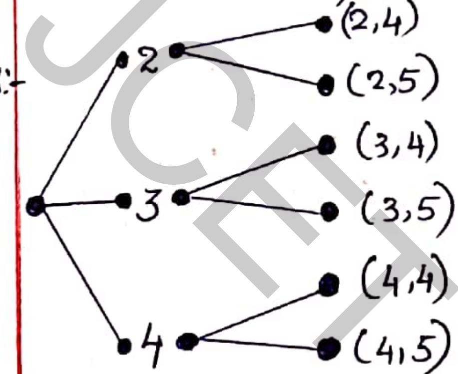
$B^3 = B \times B \times B = \{(x,x,x), (x,x,y), (x,y,x), (x,y,y), (y,x,x), (y,x,y), (y,y,x), (y,y,y)\}$

TREE DIAGRAM:- The pictorial representation of $A \times B$

Eg:- $A = \{2, 3, 4\}$ $B = \{4, 5\}$ $C = \{x, y\}$

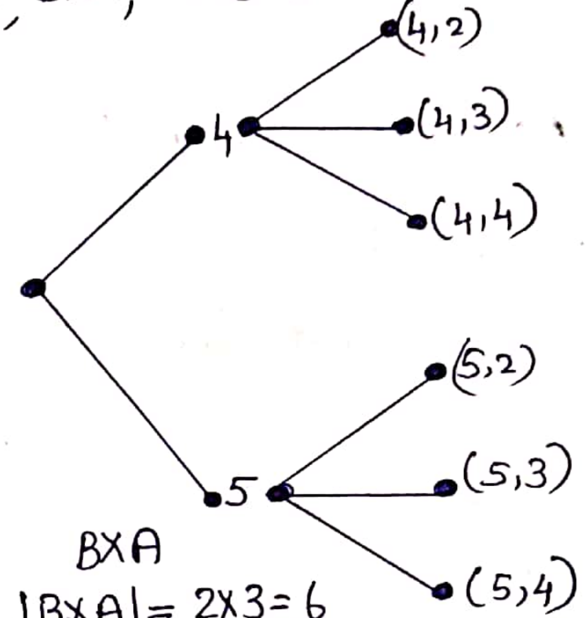
Draw tree diagrams of $A \times B$, $B \times A$, $A \times B \times C$ and find $|A \times B|$, $|B \times A|$, $|A \times B \times C|$

Ans:-



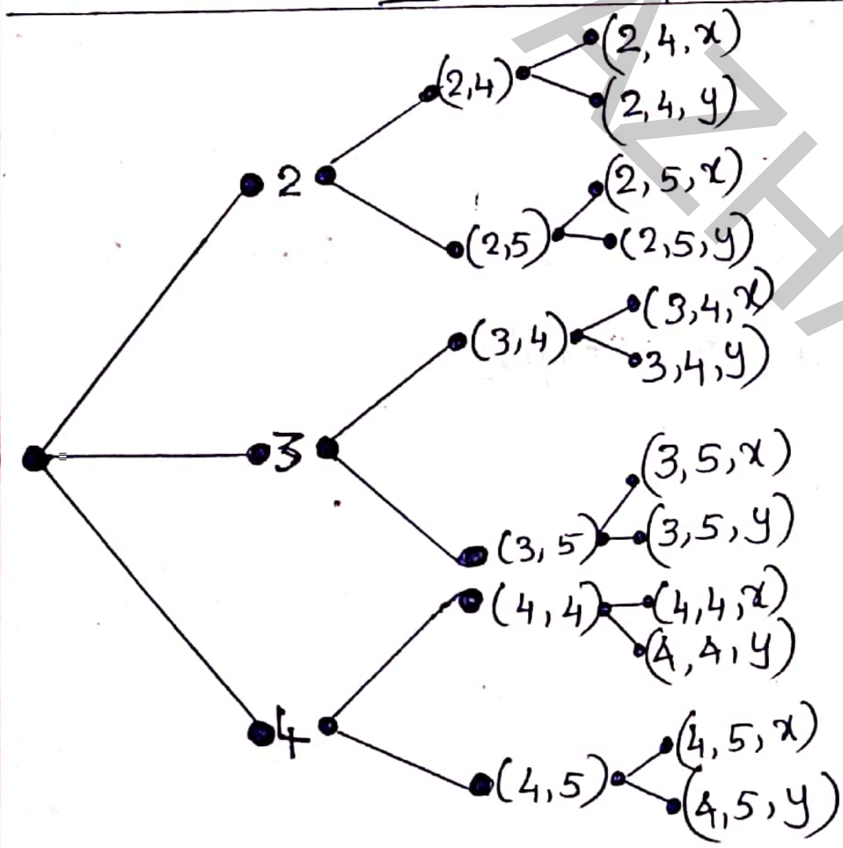
$A \times B$

$|A \times B| = 3 \times 2 = \underline{\underline{6}}$



$B \times A$

$|B \times A| = 2 \times 3 = \underline{\underline{6}}$



$A \times B \times C$.

$|A \times B \times C| = 3 \times 2 \times 2$
$= \underline{\underline{12}}$

BINARY RELATION (RELATION)

For sets A, B any subset of $A \times B$ is called a binary (Relation) from A to B.
Any subset of $A \times A$ is called a relation on A.

Note:- Since we deal with binary relations, the word "relation" will mean binary relation, unless something otherwise is specified.

Total number of relations from (A to B) & (B to A)

If $|A| = m$, $|B| = n$, then total number of relations $= \boxed{2^{mn}}$ (for both $A \to B$ & $B \to A$) $\left(\because 2^{mn} = 2^{nm}\right)$

Notation $\mathcal{P}$

If $A = \{1, 2, 3\}$ then $\mathcal{P}(A)$ denotes the set of all subsets of A

ie $\mathcal{P}(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{1,2\}\ \{1,3\}, \{2,3\}, \{1,2,3\}\}$ are the possible subsets. ($2^3 = 8$ subsets).

$$\boxed{\text{If } |A| = n \text{ then } |\mathcal{P}(A)| = 2^n.}$$

RESULT:-

For any sets $A, B, C \subseteq \mathcal{U}$

a) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

b) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

c) $(A \cap B) \times C = (A \times C) \cap (B \times C)$

d) $(A \cup B) \times C = (A \times C) \cup (B \times C)$

PROBLEMS.

1) Let $A = \{a, b, c, d\}$ $B = \{1, 2, 3\}$

a) Give examples ~~for~~ of 3 non empty relations from A to B

b) 3 examples of nonempty relations on A.

Ans (a) (i) Relation is a subset of $A \times B$.

$\{(a,1), (a,2), (a,3)\}$

(ii) $\{(b,2), (b,3), (c,2), (d,3)\}$

(iii) $\{(d,3), (c,1), (b,3), (d,1), (a,1), (a,2)\}$ etc

(b) (i) $\{(a,b), (a,c), (a,d)\}$

(ii) $\{(a,a)\}$

(iii) $\{(d,a), (d,c), (d,b), (c,b), (b,c), (a,d), (d,a)\}$

2) Let $A = \{a, b, c, d, e, f, g, h\}$, $B = \{1, 2, 3, 4, 5\}$

How many elements are there in $P(A\times B)$.

Ans:- $A\times B = \{(a,1), (a,2), \dots (h,5)\}$

$|A\times B| = |A| \times |B| = 8\times 5 = 40$

$\therefore |P(A\times B)| = \underline{\underline{2^{40}}}$ is the number of subsets of $A\times B$. ($P(A\times B)$

3) If $A = \{1,2,3,4\}$

H.W $B = \{2, 5\}$

$C = \{3, 4, 7\}$. Determine

(i) $A\times B$

(ii) $B\times A$

(iii) $A\cup(B\times C)$ Don't use the formula.

(iv) $(A\cup B)\times C$

(v) $(A\times C)\cup(B\times C)$

Ans (i) $A\times B = \{(1,2)(1,5), (2,2), (2,5), (3,2), (3,5), (4,2), (4,5)\}$

(ii) $B\times A = \{(2,1), (2,2), (2,3), (2,4), (5,1), (5,2), (5,3), (5,4)\}$

(iii) $A\cup(B\times C) = \{1, 2, 3, 4, (2,3), (2,4), (2,7), (5,3), (5,4), (5,7)\}$

(iv) $A\cup B = \{1, 2, 3, 4, 5\}$ $C = \{3, 4, 7\}$

$(A\cup B)\times C = \{(1,3), (1,4), (1,7), (2,3), (2,4), (2,7), (3,3), (3,4)$
$(3,7), (4,3), (4,4), (4,7), (5,3), (5,4), (5,7)\}$

(v) $A\times C = \{(1,3), (1,4), (1,7), (2,3), (2,4), (2,7), (3,3), (3,4), (3,7)$
$(4,3), (4,4), (4,7)\}$

$B\times C = \{(2,3), (2,4), (2,7), (5,3), (5,4), (5,7)\}$

$(A\times C)\cup(B\times C) = \{(1,3), (1,4), (1,7), (2,3), (2,4), (2,7), (3,3), (3,4), (3,7)$
$(4,3), (4,4), (4,7), (5,3), (5,4), (5,7)\}$

4) If $A = \{1,2,3\}$ $B = \{2,4,5\}$ give examples of

H.W

(a) 3 nonempty relations from A to B.

(b) 3 " " on A

(c) $|A \times B|$

(d) The number of relations from A to B

(e) The number of relations on A

Ans (a) (i) $\{(1,2),(1,4)\}$

(ii) $\{(2,4),(2,2),(2,5),(3,5)\}$

(iii) $\{(3,2),(3,4),(3,5),(1,2),(1,4)\}$

give any examples

(b) (i) $\{(1,1)\}$

(ii) $\{(1,1),(1,2),(1,3),(3,1)\}$

(iii) $\{(2,1),(2,2),(3,1),(3,2),(3,3)\}$

$A: \{1,2,3\}$
$A: \{1,2,3\}$
$A \times A$

(c) $|A \times B| = |A| \cdot |B| = 3 \times 3 = 9$

(d) No of relations from A to B $= 2^{|A||B|} = 2^{3\times3} = \underline{\underline{2^9}}$

(e) No of relations ~~from~~ on A $(A \to A) = 2^{3\times3} = \underline{\underline{2^9}}$

$2^{|A||A|} =$

5) If $A = \{1,2,3,4,5\}$ and $B = \{\omega, x, y, z\}$. How many elements are there in $\mathcal{P}(A \times B)$

Ans:- $|\mathcal{P}(A\times B)| = 2^{|A\times B|} = 2^{5\times 4}$ $|A\times B| = |A|\,|B|$
$= 5\times 4 = 20$

$$= \underline{\underline{2^{20}}}$$

6) Let A, B be sets with $|B| = 3$. If there are 4096
H.W relations from A to B. what is $|A|$?

Ans:- Let $R: A \to B$

no of relations $= 2^{mn}$ where $|A| = m$ $|B| = n$.

$4096 = 2^{m3}$ given $|B| = 3 = n$.

find m.

$4096 = 2^{12}$

$4096 = 2^{12} = 2^{3m}$

$12 = 3m$ $\therefore m = \frac{12}{3} = \underline{\underline{4}}$

2	4096
2	2048
2	1024
2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
	2

$\therefore |A| = \underline{\underline{4}}$

II. FUNCTIONS (5.2)

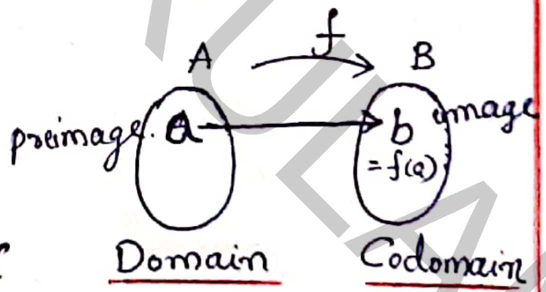
For non empty sets A, B a function or mapping 'f' from A to B denoted by $f: A \to B$ is a relation from A to B in which every element of A appears exactly once as the first component of an ordered pair in the relation.

In other words, a function $f: A \to B$ is an assignment of exactly one element of B to every element of A.

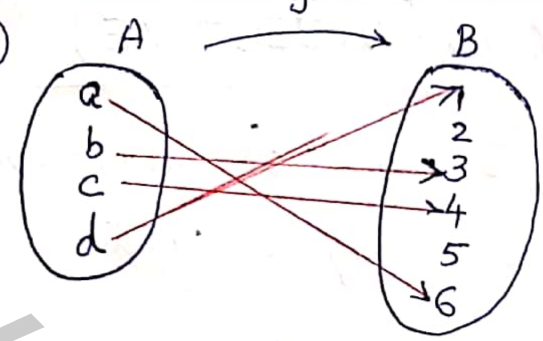
Note (Image, preimage, Domain, Codomain, Range)

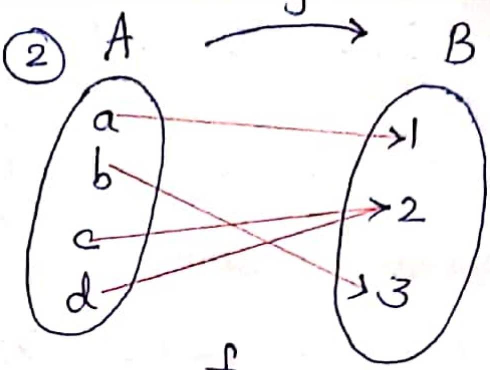
* $f(a) = b$ when (a, b) is an ordered pair in f.
  b is called the image of a under f
  a is the preimage of b under f

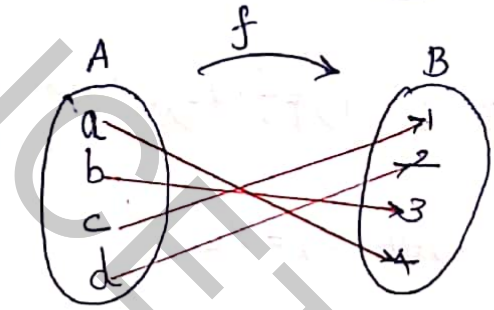
* Let $f: A \to B$
  A is called the domain of f
  B is called the co-domain of f

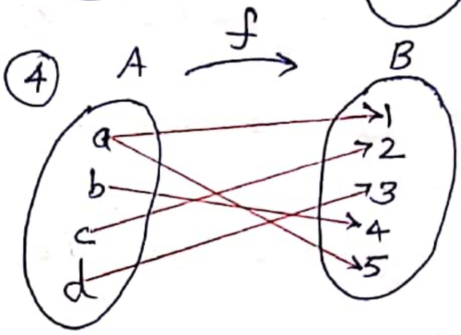


* The set of all images of A is called Range.

① 

② 

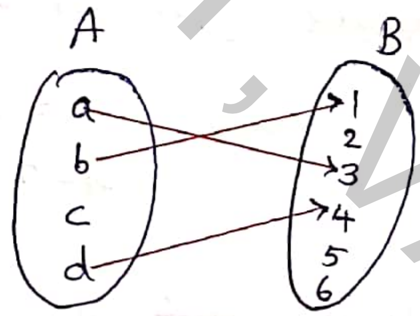
③ 

④ 

Not a function, because a has 2 images

ie $f(a)=1$, $f(a)=5$

Not possible

⑤ 

Not a function, because c has no image

Let $f: A \rightarrow B$ be a function. with $|A| = m$, $|B| = n$

Total $\boxed{\text{number of functions from } A \rightarrow B \text{ is } |B|^{|A|}} = n^m$

ONE TO ONE FUNCTIONS (INJECTIVE)

$f: A \to B$ is called one to one or injective if distinct elements of A are mapped into distinct elements of B.

ie f is one to one iff $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$

OR

$f(x_1) = f(x_2)$ whenever $x_1 = x_2$.

The number of one to one functions from A to B is $^nP_m = {}^{|B|}P_{|A|} = \dfrac{n!}{(n-m)!}$ where $|A| = m$, $|B| = n$

<u>Note</u>:- $f: A \to B$ is one to one, then $|A| \leq |B|$

Eg. In page no: 9 Egs ① & ③ are one - one functions.

1) Check whether the following function $f: R \to R$ where $f(x) = 3x+7$ $\forall x \in R$ is one to one.

Ans: Let $f(x_1) = f(x_2)$

$3x_1 + 7 = 3x_2 + 7$

$3x_1 = 3x_2$

$x_1 = x_2$

$\therefore f$ is one to one.

② $g: R \to R$ $g(x) = x^4 - x$

Ans:- $g(x_1) = g(x_2)$

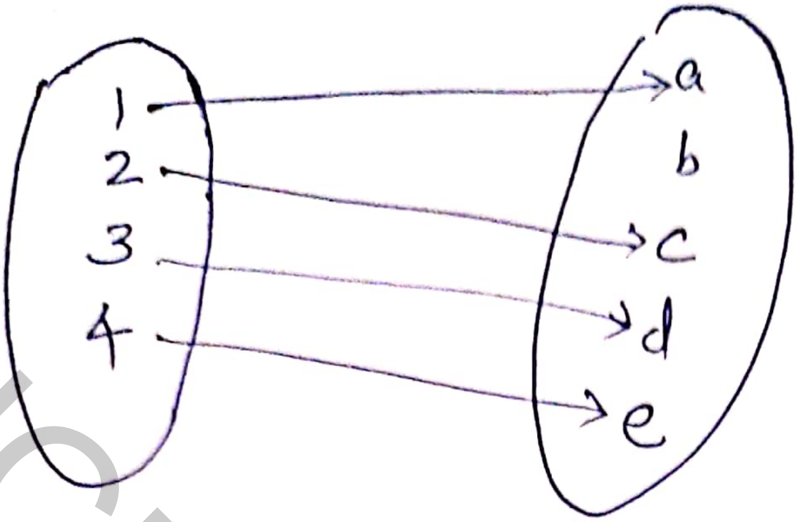
$x_1^4 - x_1 = x_2^4 - x$

Let $g(0) = 0$, $g(1) = 1$
$0^2 - 0 = 0$ $1^2 - 1 = 0$

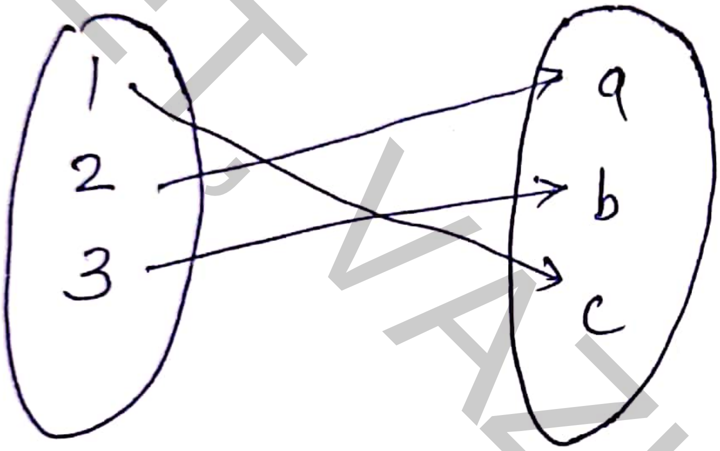
but $0 \neq 1$

ie g is not one to one

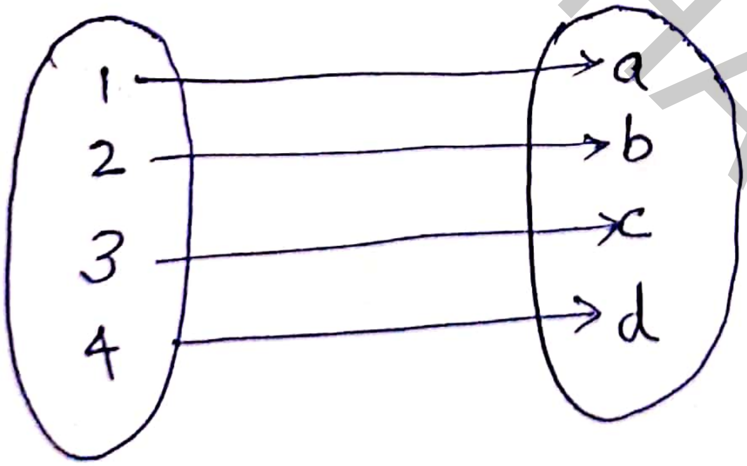
ie $\exists$ x_1, x_2 where $g(x_1) = g(x_2)$ but $x_1 \neq x_2$



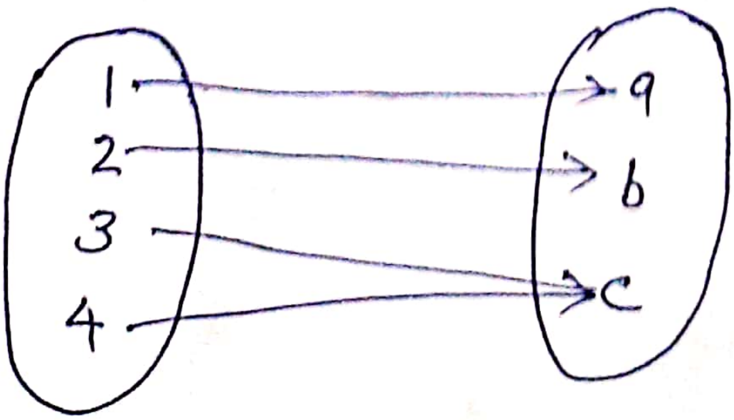
one one



one one



one one

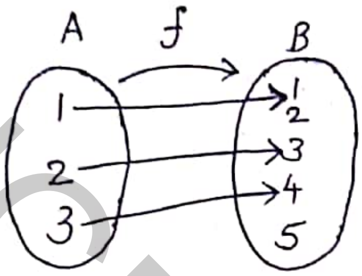


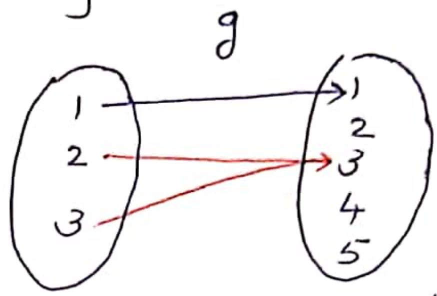
not one one

3. Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4, 5\}$ Check whether the functions $f : \{(1,1), (2,3), (3,4)\}$

$g : \{(1,1), (2,3), (3,3)\}$ are one to one

Ans:-





f is one to one

Every element in A has different images in B

g is not one to one

because $g(2) = g(3)$ but $2 \neq 3$.

2 & 3 have same image 3 in B.

$\therefore$ not one to one

4. How many functions are there from $f : A \rightarrow B$ & how many of them are one to one. (A & B are given above in Q.3)

Ans:- Total number of functions $= |B|^{|A|} = \underline{5^3} = \underline{125}$

Number of one to one functions $= {}^{|B|}P_{|A|} = {}^{5}P_{3}$

$$= \frac{5!}{(5-3)!} = \frac{5!}{2!}$$

$$= \frac{5 \times 4 \times 3 \times 2 \times 1}{1 \cdot 2}$$

$$= \underline{60}$$

~~Definition~~

5) Determine which of the following functions are one to one and find its range.

(a) $f: Z \to Z : f(x) = 2x$

(b) $f: Q \to Q : f(x) = 2x$

(c) $f: R \to R : f(x) = e^{x^2}$

(d) $f: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to R : f(x) = \cos x$

H.W

(e) $f: Z \to Z,\ f(x) = 2x+1$

(f) $f: Q \to Q,\ f(x) = 2x+1$

(g) $f: Z \to Z,\ f(x) = x^3 - x$

(h) $f: R \to R,\ f(x) = e^x$

(i) $f: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to R \quad f(x) = \sin x$

(j) $f: [0, \pi] \to R \quad f(x) = \sin x$

Std NOTATIONS

Z : Set of integers $\{\ldots -4, -3, -2, -1, 0, 1, 2, \ldots\}$

Z^+ : $\{1, 2, 3, 4, \ldots\}$ +ve Integers

Z^- : $\{\ldots -4, -3, -2, -1\}$ –ve integers

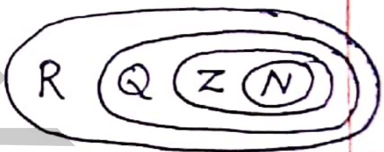
Q : Set of Rational numbers

fractions $\{\frac{4}{7}, 1, \frac{4}{3}, 12\frac{7}{16}, 202, \frac{31}{129} \ldots\}$

$\{\frac{p}{q} \mid p, q \in Z,\ q \neq 0\}$

R : Set of all real numbers $\{-1, -\frac{1}{2}, \sqrt{2}, \pi, 9 \ldots\}$

N : Set of Natural numbers $\{1, 2, 3, 4, \ldots\}$ Counting numbers



Ans (a) $f(x_1) = f(x_2)$

$2x_1 = 2x_2$

$x_1 = x_2$

ie $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

$\therefore$ (a) is a one to one function.

Range $= \{2x / x \in Z\}$ means set of even numbers.

$= \{2, 4, 6, 8, \ldots\}$

(b) is a one to one function.

Range $= \{2x / x \in Q\} = Q$.

(c) $x_1 = -1, \quad x_2 = +1$

$f(x_1) = e^{x_1^2} = e^{(-1)^2} = e, \quad f(x_2) = e^{x_2^2} = e^{1^2} = e$

ie $x_1 \neq x_2 \Rightarrow f(x_1) = f(x_2)$

$\therefore$ not one to one.

Range $= \{[0, \infty)\}$

(d) $x_1 = -\frac{\pi}{2}$ $\qquad$ $x_2 = \frac{\pi}{2}$

$f(x_1) = \cos(-\frac{\pi}{2})$ $\qquad$ $f(x_2) = \cos\frac{\pi}{2}$

$= \cos\frac{\pi}{2} = 0$ $\qquad$ $\cos\frac{\pi}{2} = 0$

ie $x_1 \neq x_2 \implies f(x_1) = f(x_2)$ $\quad \therefore$ not one to one

Range $= \underline{\underline{[0, 1]}}$

(d) $x_1 = -\frac{\pi}{2}$ $\quad$ $x_2 = \frac{\pi}{2}$

$f(x_1) = \cos(-\frac{\pi}{2})$ $\quad$ $f(x_2) = \cos\frac{\pi}{2}$

$= \cos\frac{\pi}{2} = 0$ $\quad$ $\cos\frac{\pi}{2} = 0$

ie $x_1 \neq x_2 \Rightarrow f(x_1) = f(x_2)$ $\quad \therefore$ not one to one

Range $= \underline{\underline{[0, 1]}}$

(e) one - one

Range $= \{(2x+1) / x \in Z\} = \{1, 3, 5, 7, \ldots\}$ set of odd integers

(f) One - One

Range $= Q$

(g) Not one - one

$f(x_1) = f(x_2)$ $\quad$ Let $x_1 = 0, x_2 = 1$

$0 = 0$

but $x_1 \neq x_2$

Range $= \{0, \pm 6, \pm 24, \pm 60, \ldots\}$

$= \{n^3 - n / n \in Z\}$

(h) one - one $\quad$ Range $= (0, +\infty)$

$f(x_1) = f(x_2)$ $\quad$ Range $= \{e^x / \in R\}$ $\quad$ exponential function is always +ve.

$e^{x_1} = e^{x_2}$

ie $x_1 = x_2$

(i) one - one. $\quad$ Range $= [-1, 1]$

(j) Not one-one

$f(x_1) = f(x_2)$ $\quad$ Range $= [0, 1]$

$f(0) = f(\pi) = 0$ $\quad$ positive values only.

but $0 \neq \pi$

Definition   **Image of a subset of A**

If $f: A \to B$ and $A_1 \subseteq A$ then

$f(A_1) = \{ b \in B / f(a) = b$ for some $a \in A_1 \}$ and

$f(A_1)$ is called image of A_1 under f.

1) $A = \{1, 2, 3, 4, 5\}$

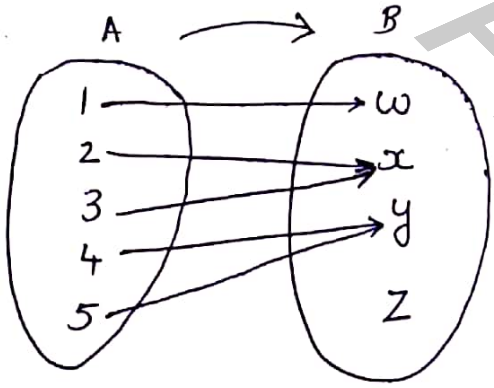
$B = \{w, x, y, z\}$ Let $f: A \to B$ be given by

$f = \{ (1, w), (2, x), (3, x), (4, y), (5, y) \}$. Let $A_1 = \{1\}$,

$A_2 = \{1, 2\}$, $A_3 = \{1, 2, 3\}$, $A_4 = \{2, 3\}$, $A_5 = \{2, 3, 4, 5\}$

Find $f(A_1)$, $f(A_2)$, $f(A_3)$, $f(A_4)$, $f(A_5)$ | Images set of A_1, A_2, A_3, A_4, A_5

Ans:-



$f(A_1) = \{ f(1) \} = \{ w \}$

$f(A_2) = \{ f(1), f(2) \} = \{ w, x \}$

$f(A_3) = \{ f(1), f(2), f(3) \} = \{ w, x \}$

$f(A_4) = \{ x \}$

$f(A_5) = \{ x, y \}$

2) $g: R \to R$ given by $g(x) = x^2$. Let $A_1 = [-2, 1]$ (closed interval)

What is the image of A_1.

Ans:- $A_1 = [-2, 1] \subset R$

$A_1 = \{ -2, \ldots -1, \ldots 0, \ldots \frac{1}{2} \ldots 1 \}$

$g(A_1) = \{ g(-2) \ldots, g(-1), \ldots g(0), \ldots g(1) \} = \{ 4, \ldots 1, \ldots 0, \ldots \frac{1}{4}, \ldots 1 \}$

$\therefore g(A_1) = [0, 4]$ closed interval.

H.W
3) Let $f: R \to R$ where $f(x) = x^2$. Determine $f(A)$ for the following.

a) $A = \{2, 3\}$ b) $A = (-3, 3)$ c) $A = [-7, 2]$

d) $A = (-3, 2]$ f) $[-4, -3] \cup [5, 6]$

4) If $A = \{1, 2, 3, 4, 5\}$ and there are 6720 injective (one-to one) functions from $f: A \to B$ what is $|B|$
H.W

Ans:- No of one to one functions $= {}^{|B|}P_{|A|} =$
Hint

$|A| = 5$
$|B| = n$

ie $^{n}P_5 = 6720$

$$\frac{n!}{(n-5)!} = 6720$$

H.W
5) If there are 2187 functions $f: A \to B$ and $|B| = 3$, what is $|A| = ?$

Hint Total number of functions from $A \to B$ is $|B|^{|A|}$

Let $|A| = m$
find m.

$2187 = 3^m$

H.W

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Ans:- No of one to one functions = $^{|B|}P_{|A|}$ =

Hint

$|A| = 5$

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ie $^nP_5 = 6720$

$\frac{n!}{(n-5)!} = 6720$

$n(n-1)(n-2)(n-3)(n-4) = 6720$

$8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = 6720$

$\therefore \underline{n = 8}$

$$\begin{array}{r|l} 2 & 6720 \\ \hline 2 & 2360 \\ \hline 2 & 1280 \\ \hline 2 & 640 \\ \hline 2 & 32 \\ \hline 2 & 16 \\ \hline & 8 \end{array}$$

H.W

5) If there are 2187 functions $f: A \to B$ and $|B| = 3$, what is $|A| = ?$

Hint Total number of functions from $A \to B$ is $|B|^{|A|}$

Let $|A| = m$

find m.

$2187 = 3^m$

$2187 = 3^7$

$\therefore \underline{\underline{m = 7}}$

$$\begin{array}{r|l} 3 & 2187 \\ \hline 3 & 729 \\ \hline 3 & 243 \\ \hline 9 & 81 \\ \hline & 9 \end{array}$$

RESULT

Let $f: A \to B$ with $A_1, A_2 \subseteq A$ then

ⓐ $f(A_1) \cup f(A_2) = f(A_1) \cup f(A_2)$

ⓑ $f[(A_1) \cap (A_2)] \subseteq f(A_1) \cap f(A_2)$

ⓒ $f(A_1 \cap A_2) = f(A_1) \cap f(A_2)$ when f is one-one.

RESTRICTION of a function f to A_1 (f/A_1)

If $f: A \to B$ where $A_1 \subseteq A$ then

$f/A_1 : A_1 \to B$ is called the restriction of f to A_1

if $f/A_1(a) = f(a)$ for all $a \in A_1$

choosing a smaller domain A_1 for the original function f

EXTENSION OF f to A

Let $A_1 \subseteq A$ and $f: A_1 \to B$.

If $g: A \to B$ and $g(a) = f(a)$ $\forall a \in A_1$, then

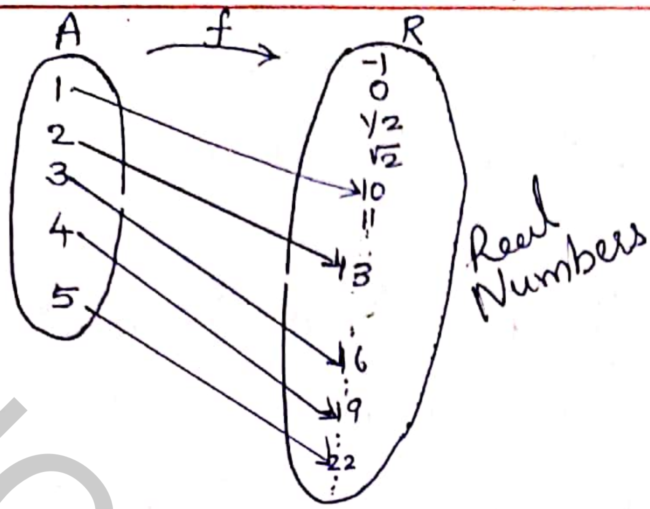
we call g an extension of f to A

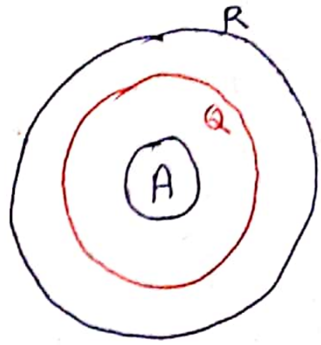
Eg:- For $A = \{1,2,3,4,5\}$ $\boxed{f: A \to R}$ defined by

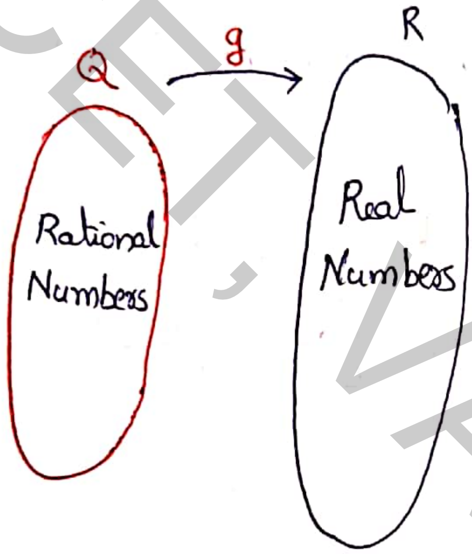
$f = \{(1,10), (2,13), (3,16), (4,19), (5,22)\}$

Let $\boxed{g: Q \to R}$ defined by $g(q) = 3q + 7$ $\forall q \in Q$.

finally $\boxed{h: R \to R}$ defined by $h(r) = 3r + 7$ $\forall r \in R$.







i) f is the restrn of g (from Q) to A

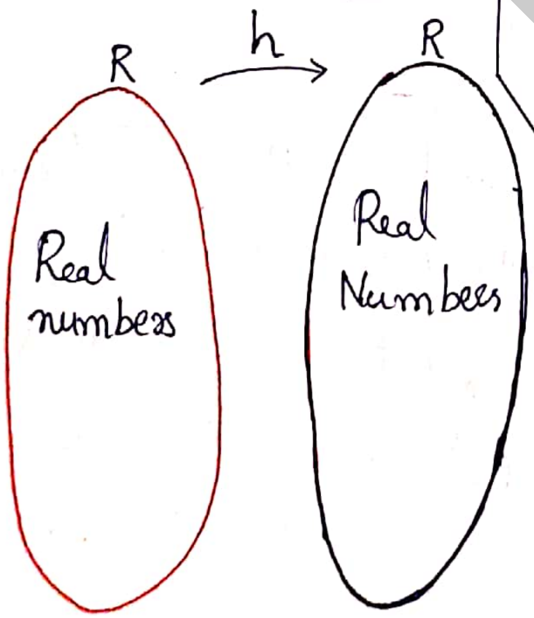
(ii) f is the restrn of h (from R) to A

(iii) h is an extension of f (from A) to R

(iv) h is an extension of g (from Q) to R

(v) g is an extension of f (from A) to Q.

(vi) g is a restriction of h (from R) to Q



Q) Determine whether or not each of the following relations are functions. If a relation is a function find its range

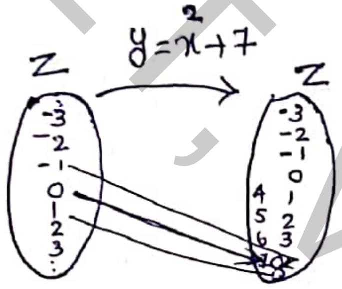
(i) $\{(x,y) / x, y \in Z,\ y = x^2+7\}$ a relation from Z to Z

(ii) $\{(x,y) / x, y \in R,\ y^2 = x\}$, a relation from R to R

H.W (iii) $\{(x,y) / x, y \in R,\ y = 3x+1\}$, a relation from R to R

H.W (iv) $\{(x,y) / x, y \in Q,\ x^2+y^2=1\}$ a relation from Q to Q

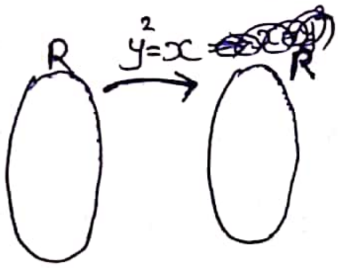
Ans:- (i)



A relation and a function

$\therefore$ Range $= \{7, 8, 11, 16, 23, \dots\}$

(ii)



$y^2 = x$

$y = \pm\sqrt{x}$

when $x = -3$

we cannot map -3 to a +ve real number.

-ve numbers have no image

$\therefore$ A relation, but not a function.

$\therefore$ No range

(iii)

H.W

Q. Check whether $f: R \to R$ where $f(x) = \frac{1}{x^2-2}$ and $f: Z \to R$ where $f(x) = \frac{1}{x^2-2}$ are functions or not

Ans:- $f: R \to R$, when $x = \sqrt{2} \in R$ (A real number)

$R \xrightarrow{f} R$

$f(x) = \frac{1}{(\sqrt{2})^2 - 2} = \frac{1}{0} \notin R$ (has no image)

$\therefore$ $f: R \to R$ is not a function under $f(x) = \frac{1}{x^2-2}$

$f: Z \to R$

$Z \xrightarrow{f} R$

$x = \{\ldots -3, -2, -1, 0, 1, 2, 3, 4, \ldots\} \in Z$

$f(x) = \{\frac{1}{x^2-2}\}$ a real number

$\therefore$ Every element in Z has exactly one image $\therefore$ A function

Range $= \{R\}$